JEREMY DO

740-915-3932 | do_g1@denison.edu | /in/do-hoang-gia-huy-bbb05921b/ | /github.com/dohoanggiahuy317

EDUCATION

Denison University Granville, OH Graduation: May 2026

- **Bachelor of Science** in Computer Science and Applied Mathematics (**Major GPA: 4.0**)
- **Coursework**: Object-Oriented Design, Data structures & Algorithms, Cloud services, Data Systems, Microservices and Serverless

SKILLS

Development Tools: C/C++, Java, JavaScript, Python, React, Assembly, Spring boot, Firebase, Git, Postman, CI/CD, CRUD, RPC
Cloud Tools: Docker, Kubernetes, MySQL, PostgreSQL, Redis, Monasca, OpenStack, AWS, Microsoft Azure, Kafka, RabbitMQ

EXPERIENCE

Deloitte May 2024 – August 2024
Software Engineer Intern *Hanoi, Vietnam*

- Utilized Jmix and Spring Boot to develop features for monitoring and securing **4,000+** bank transactions and contracts per day.
- Reduced loading time by **2 min** by optimizing queries in the Persistence layer for Oracle SQL Database using aggregation pipelines.
- Designed 4-cluster Kubernetes to enhance a search API to handle **500** transactions, optimizing server traffic by **25%.**
- Ensured secure transactions between Kafka brokers and clients by implementing SSL/TLS encryption and SASL authentication.

FPT Software July 2023 – September 2023
Software Engineer Intern *Hanoi, Vietnam*

- Designed a soft deletion feature to facilitate data center procedure correction using C#, Amazon RDS, and DynamoDB.
- Upgraded the hypervisor manager using Monasca, reducing computing costs by **65%** for **85** cloud-based virtual machines.
- Utilized RabbitMQ for async processing and Redis for caching to handle clients' requests, reducing the response time by 3 sec.
- Increased scalability by **34%** using Docker with a 3-process container to optimize data flow among servers and clients.

VinBrain x Microsoft Smart Health Lab May 2022 – September 2022
Software Engineer Intern *Hanoi, Vietnam*

- Developed a Spring Boot microservice booking app to support **100+** patients and doctors during the COVID-19 pandemic.
- Built a client tracking system on Azure Functions from Papertrail logs, improving client real-time updates speed by **2 times**.
- Facilitated seamless doctor actions across Node.js multi-services by implementing distributed tracing.
- Enhanced security at the API Gateway by utilizing HS256 JSON Web Token encryption and synchronous communication.

Above Data, Inc January 2024 – April 2024
Software Engineer Intern *Granville, OH*

- Designed a RAG pipeline with LLMs to generate natural responses to customer queries about information on **500+** retail brands.
- Improved data retrieval by 33% by designing business logic for RESTful API between the OLlama server and database in Docker.

Denison University Research Lab May 2023 – July 2023
Data Engineer Intern *Granville, OH*

- Proposed a multi-threaded pipeline to generate a high-quality dataset of **77,000** data entries for LLMs model training.
- Developed a RESTful API to abstract the data flows among microservices. Link to paper: https://arxiv.org/abs/2309.05103v1

PROJECTS

Student Teamwork Management System

- Created a light SQL engine with distributed Parallel Buffer Pool for managing **3,000** teamwork assignments at my college.
- Designed a scalable Query Execution Engine capable of parallel query processing and distributed computing using C++.
- Optimized cache replacement policies to achieve **O(1)** time complexity using Linked Lists, Hash Tables, and B+ Tree Indexing.

TikTok TechJam Hackathon 2024 - Image 2 LaTex converter

- Built a full-stack app to convert images to LaTeX on AWS S3 to reduce learning stress for **500** students in my department.
- Reduced Flask backend memory usage by **1GB** by designing lazy loading method for model inference and image preprocessing.

Cornell University Hackathon 2020

- Built an autonomous software system utilizing OpenCV and created 3D printed prototype of self-driving car using AutoCAD.
- Converted the sensor's input data to grayscale to prevent overheating when transmitting data to the Arduino microcontroller.

AWARDS

National Mathematics Olympiad 2019: Third Prize in the Vietnam National Mathematics Olympiad
Ohio Annual Programming Contest 2023 & 2024: First Prize among 45 programming teams from 4 colleges in Ohio.
Extracurriculars: Microsoft Learn Student Ambassador, Tapia '23, Denison CompSci Club